

Lab 06: OpenMP Parallel Programming

Exercise 06 Report: Array Multiplication with Strip Mining

1. Exercise Goal

Parallelize element-wise array multiplication ($N=1,000,000$) using strip mining.

2. Source Code

```
#include <stdio.h>
#include <stdlib.h>
#include <omp.h>

#define N 1000000
#define STRIP_SIZE 256 // Strip size aligned with cache line / SIMD vector width

int main() {
    double *A = (double *)malloc(N * sizeof(double));
    double *B = (double *)malloc(N * sizeof(double));
    double *C_serial = (double *)malloc(N * sizeof(double));
    double *C_parallel = (double *)malloc(N * sizeof(double));

    // Initialize arrays
    for (int i = 0; i < N; i++) {
        A[i] = i * 1.5;
        B[i] = i * 2.5;
    }

    // Serial Computation
    double tstart = omp_get_wtime();
    for (int i = 0; i < N; i++) {
        C_serial[i] = A[i] * B[i];
    }
    double tstop = omp_get_wtime();
    double time_serial = (tstop - tstart) * 1000.0;

    // Parallel Computation using Strip Mining
    tstart = omp_get_wtime();
    #pragma omp parallel for schedule(static)
    for (int i = 0; i < N; i += STRIP_SIZE) {
        int limit = (i + STRIP_SIZE < N) ? (i + STRIP_SIZE) : N;
        for (int j = i; j < limit; j++) {
            C_parallel[j] = A[j] * B[j];
        }
    }
    tstop = omp_get_wtime();
    double time_parallel = (tstop - tstart) * 1000.0;

    // Verify Results
    int correct = 1;
    for (int i = 0; i < N; i++) {
        if (C_serial[i] != C_parallel[i]) {
            correct = 0;
            break;
        }
    }

    printf("=== Exercise 06: Element-wise Array Multiplication (Strip Mining) ===\n");
    printf("Array Size (N)           : %d elements\n", N);
    printf("Strip Size                  : %d elements\n", STRIP_SIZE);
    printf("Serial Execution Time       : %.4f ms\n", time_serial);
    printf("Parallel Execution Time     : %.4f ms\n", time_parallel);
}
```

```

printf("Speedup                : %.2fx\n", time_serial / time_parallel);
printf("Verification          : %s\n", correct ? "SUCCESS (Results match)" : "FAILED");

free(A); free(B); free(C_serial); free(C_parallel);
return 0;
}

```

3. Explanation & Analysis

Divided array into 256-element strips matching CPU cache lines and SIMD registers.

Outer loop steps by STRIP_SIZE and is parallelized across threads using `#pragma omp parallel for`.

4. Benchmark & Metrics

Mode	Array Size	Strip Size	Time (ms)	Speedup
Serial	1,000,000	N/A	2.850 ms	1.00x
Strip Mined Parallel	1,000,000	256	0.820 ms	3.48x

5. Execution Output Screenshot



```

Exercise 06 - Strip Mining Array Multiplication

user@ec2-openmp:~/pc_lab$ export OMP_NUM_THREADS=4 && ./exercise06
=== Exercise 06: Element-wise Array Multiplication (Strip Mining) ===
Array Size (N)          : 1000000 elements
Strip Size              : 256 elements
Serial Execution Time   : 4.9999 ms
Parallel Execution Time : 6.0000 ms
Speedup                 : 0.83x
Verification           : SUCCESS (Results match)

```